

WILLIAM CHIU

416-568-2618 | w4chiu@uwaterloo.ca | [linkedin.com/in/williamrchiu](https://www.linkedin.com/in/williamrchiu) | github.com/WilliamRChiu | williams-portfolio-nu.vercel.app/

Education

University of Waterloo

2029

Bachelor of Software Engineering, Honours Co-Op

Waterloo, ON | CGPA: 3.93/4.00

Waterloo Engineering Endowment Fund Representative — 1A & 1B

Colonel Hugh Heasley Scholarship Recipient

Work Experience

Blair AI

January 2026 – April 2026

Software Engineer

Toronto, ON

- Enabled clinic-wide support teams to manage real-time SMS conversations with patients directly in Slack threads, by building an AI agent tool-call workflow using the **Slack API**, **Twilio**, **NestJS**, and **Supabase** subscriptions.
- Saved **7+ staff hours per week** in manual fax-filing work by building a multimodal AI fax classification platform designed to process **60,000+ documents annually**, with normalized **Postgres** schema design, **OCR-oriented preprocessing** for rotation correction and text sharpening, **Avaros EMR** integration, and Terraform-managed **AWS** infrastructure across **S3**, **ECS/Fargate**, and **CloudWatch**.
- Improved **OCR prompt performance** by **35%** across **1,000+ labeled faxes** by building an **LLM-driven prompt refinement and benchmarking platform** with **concurrent evaluation**, **automated labeled-data ingestion**, **stratified sampling**, **metric summarization**, and **historical backtesting**.
- Won **6-figure annual contracts** from major hospitals by extending the fax-processing platform into an end-to-end **MRI referral system** with a scalable multi-tenant **Postgres** model, Terraform-managed **AWS SES/SNS** communication pipelines, and **voice AI** integration for autonomous follow-up calls

Ontario Government (MPBSDP - GSIC)

May 2025 – August 2025

Full Stack Developer

Toronto, ON

- Modernized **CI/CD** with a **Kubernetes**-based architecture and templated **Azure DevOps** pipelines, achieving **zero-downtime deployments** for the organ donor registration platform.
- Led development of the Organ Donor Registration backend, improving system reliability through a **Java Spring Boot** architecture, **C#** API Manager integrations, **OpenAPI/Swagger**-documented **REST APIs**, and a **WebClient circuit breaker** for failed API calls.
- Reduced email template retrieval latency by **80%** and protected sensitive data for **16M+ Ontarians** by implementing **Caffeine caching**, scalable functional interface-based validation, and an **AOP logging** framework with a **DFS JSON** data masker for **Azure Application Insights**.

Wat.AI: SEE-DR

September 2025 – March 2026

Deployment and Modeling Team

Waterloo, ON

- Evaluated **Swin-UNet** and **CMAC-UNet** for diabetic retinopathy lesion segmentation through model comparison
- Improved lesion segmentation robustness to class imbalance by developing a composite **Tversky+Dice** loss

Projects

Polyhedral Order Book | C++20, CMake, Catch2, HiGHS

- Implemented a **polyhedral order book** in C++20, generalizing order crossing to N assets via **primal/dual LPs** solved with HiGHS, enabling **multi-leg trades** and **correlated-basket market making**.
- Validated correctness across **crossed/uncrossed books**, **partial fills**, and **multi-level trades** against a baseline **2-asset CLOB** with a **Catch2** test suite under **CMake/Ninja**.

Quantum Computing Algorithm to Minimize Power Loss | Python, D-Wave

- Minimized power loss in energy systems** through a cloud-based quantum optimization program by designing the **DFS-based** search strategy and deriving the **edge-weighting formula** for the optimization algorithm.
- Recognized with a **National Silver Medal**, **Energy Challenge Award**, and **Hydrogen-Optimized Award**.

Other Projects | HTML, CSS, JavaScript, Python, Gemini, 3JS, FastAPI, AssemblyAI, Blender, Vercel, Bash

- Developed a voice-controlled **Gemini Chrome Extension** to help visually impaired users navigate the web.
- Building a bug-killing robot with **OpenCV** insect detection, **A*** path planning, and **closed-loop pursuit** logic.
- Engineered a voice-controlled wheelchair prototype powered by a **Raspberry Pi Zero**.

Technical Skills

Languages: C/C++, Java, Python, Javascript, TypeScript, HTML/CSS, SQL, Scala, VHDL

Libraries/Frameworks: React, Node.js, Express, Next.js, FastAPI, SpringBoot, NestJS, 3JS

Developer Technologies: Git, SupaBase, Postman, Docker, Kubernetes, Azure, AWS, Swagger, Vercel, Railway